MDTS4214_Assignment_7_710

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Problem Set 7: Shrinkage Methods

We would be working with the Hitters dataset available in the ISLR library. We have to predict the Salary based on all available variables. First, we make sure that we delete all rows with one or more missing values.

1. We split the data set into a training set and a test set randomly, keeping 100 observations in the test set and the remaining in the training set.

```
## install.packages("glmnet")

library(ISLR)

## Warning: package 'ISLR' was built under R version 4.5.2

library(glmnet)

## Warning: package 'glmnet' was built under R version 4.5.3

## Loading required package: Matrix

## Loaded glmnet 4.1-10

data(Hitters)

# Remove missing values
Hitters = na.omit(Hitters)

dim(Hitters)

## [1] 263  20

set.seed(123)

test_index = sample(1:nrow(Hitters), 100)

test_data = Hitters[test_index, ]
train_data = Hitters[-test_index, ]

dim(train_data)

## [1] 163  20

dim(test_data)

## [1] 100  20
```

```r
# Create model matrices (needed for glmnet)
x_train = model.matrix(Salary ~ ., train_data)[, -1]
y_train = train_data$Salary

x_test = model.matrix(Salary ~ ., test_data)[, -1]
y_test = test_data$Salary
```

2. We train the model using least squares and report the train MSE and test MSE.

```r
lm_model = lm(Salary ~ ., data = train_data)
summary(lm_model)

##
## Call:
## lm(formula = Salary ~ ., data = train_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -809.76 -182.13  -54.69  138.36 1837.14
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  212.67449  129.87001   1.638 0.103705
## AtBat         -2.04895    0.87954  -2.330 0.021229 *
## Hits           5.85495    3.15579   1.855 0.065612 .
## HmRun          5.43622    9.22155   0.590 0.556448
## Runs          -2.49915    4.03131  -0.620 0.536288
## RBI            2.32756    3.84573   0.605 0.545984
## Walks          6.40986    2.41415   2.655 0.008827 **
## Years         -8.30051   17.14327  -0.484 0.628996
## CAtBat         0.01569    0.19354   0.081 0.935512
## CHits         -0.54632    1.02039  -0.535 0.593204
## CHmRun        -2.38189    2.49628  -0.954 0.341604
## CRuns          1.56320    1.05024   1.488 0.138840
## CRBI           1.26993    1.02979   1.233 0.219524
## CWalks        -0.95946    0.45825  -2.094 0.038049 *
## LeagueN       44.32184   98.13766   0.452 0.652220
## DivisionW   -149.54532   56.87996  -2.629 0.009496 **
## PutOuts        0.41589    0.11170   3.723 0.000282 ***
## Assists        0.43589    0.28787   1.514 0.132176
## Errors        -5.50162    5.83866  -0.942 0.347642
## NewLeagueN   -15.79562   97.46148  -0.162 0.871479
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 346.5 on 143 degrees of freedom
## Multiple R-squared:  0.4782, Adjusted R-squared:  0.4088
## F-statistic: 6.897 on 19 and 143 DF,  p-value: 1.144e-12

# Predictions
# Predictions
```

```r
train_pred_lm = predict(lm_model, newdata = train_data)
test_pred_lm  = predict(lm_model, newdata = test_data)

# MSE function
mse = function(y, yhat){
  mean((y - yhat)^2, na.rm = TRUE)
}

# Compute MSE
train_mse_lm = mse(train_data$Salary, train_pred_lm)
test_mse_lm  = mse(test_data$Salary, test_pred_lm)

train_mse_lm

## [1] 105334.2

test_mse_lm

## [1] 87803.91
```

3.We fit ridge regression models on the training set, with grid values of λ. We report the following:

(i) the dimension of the coefficient matrix so obtained.

(ii) The l2 norm of the regression coefficients corresponding to the 50th and 60th choice of λ. We comment on the output.

(iii) Test MSE corresponding to these two choices of λ.

```r
lambda_grid = 10^seq(10, -2, length = 100)

ridge_model = glmnet(x_train, y_train, alpha = 0, lambda = lambda_grid)
```

(i) Dimension of Coefficient Matrix

```r
dim(coef(ridge_model))

## [1]  20 100
```

Coefficients for 50th and 60th λ

```r
coef_50 = coef(ridge_model)[, 50]
coef_60 = coef(ridge_model)[, 60]

coef_50

##   (Intercept)         AtBat          Hits         HmRun          Runs
## 438.509969440   0.029684297   0.110523230   0.560867468   0.178502584
##           RBI         Walks         Years        CAtBat         CHits
##   0.252163513   0.238337160   1.002466622   0.002796464   0.010194612
##        CHmRun         CRuns          CRBI        CWalks       LeagueN
##   0.077846378   0.020089803   0.021957346   0.021866228   0.432252998
##     DivisionW       PutOuts       Assists        Errors    NewLeagueN
##  -6.916280599   0.018294166   0.002998934  -0.082552703   0.741567242
```

```
coef_60

##  (Intercept)        AtBat         Hits        HmRun         Runs
RBI
## 105.63996577   0.08310078   0.51246876   2.09635144   0.60993278   1.14623
958
##        Walks        Years       CAtBat        CHits       CHmRun        CR
uns
##   1.01677160   2.77366963   0.01043635   0.04276159   0.27032634   0.08012
067
##         CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assi
sts
##   0.09127173   0.05840151  18.35350369 -62.46410535   0.14784491   0.04002
532
##       Errors   NewLeagueN
##  -0.86723778  12.21693330
```

(ii). L2 Norm of Coefficients

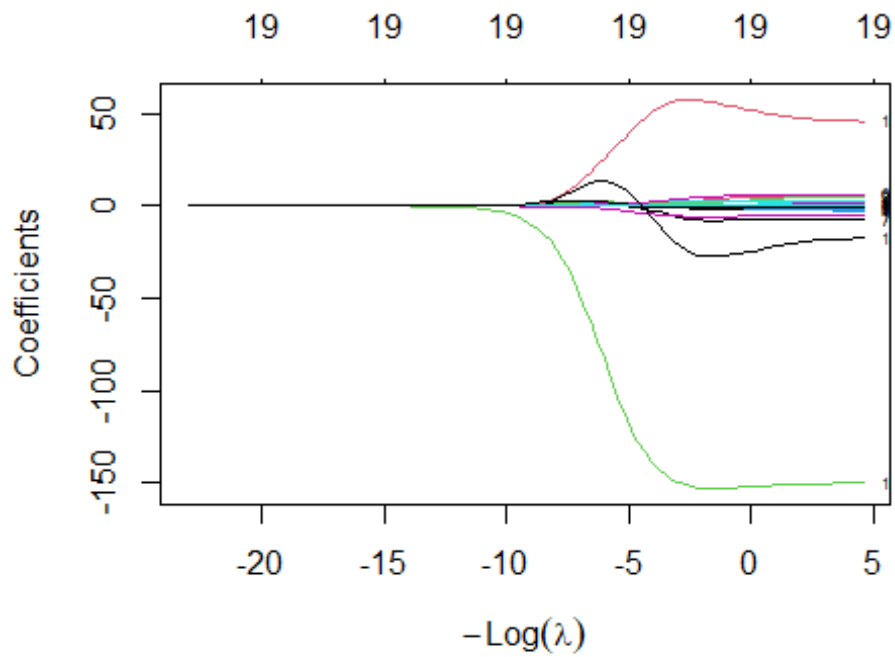
```
l2_norm = function(beta){
  sqrt(sum(beta^2))
}

l2_50 = l2_norm(coef_50)
l2_60 = l2_norm(coef_60)

l2_50

## [1] 438.5671

l2_60

## [1] 124.7542
```

(iii). Test MSE for 50th and 60th λ

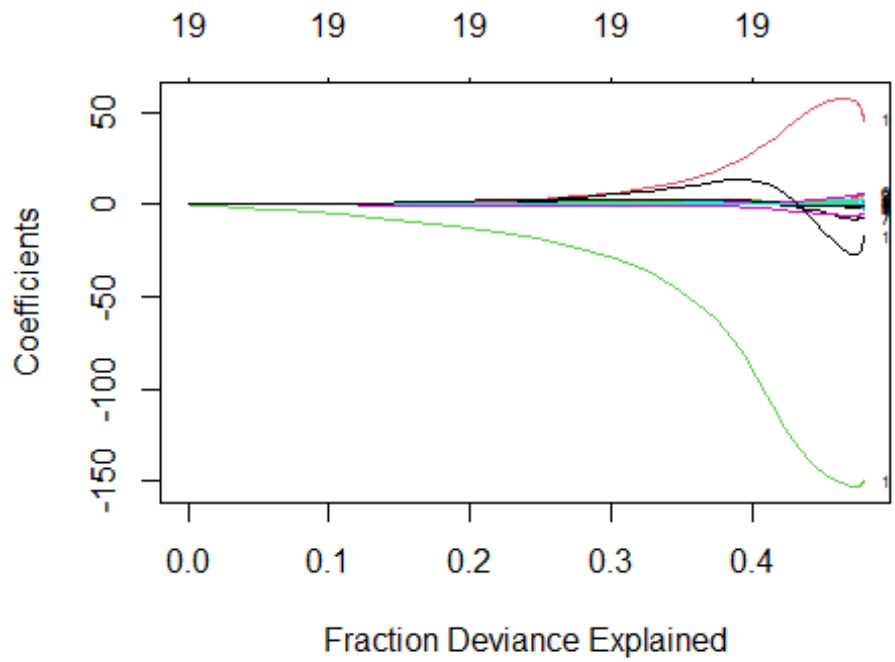
```
# Predictions
ridge_pred_50 = predict(ridge_model, s = lambda_grid[50], newx = x_test)
ridge_pred_60 = predict(ridge_model, s = lambda_grid[60], newx = x_test)

# Test MSE
mse_50 = mse(y_test, ridge_pred_50)
mse_60 = mse(y_test, ridge_pred_60)

mse_50

## [1] 167303.1

mse_60

## [1] 97935.56
```

4. Make a line plot (for all the predictors) of the standardized coefficients against the grid value of λ chosen by you. Interpret the graph.

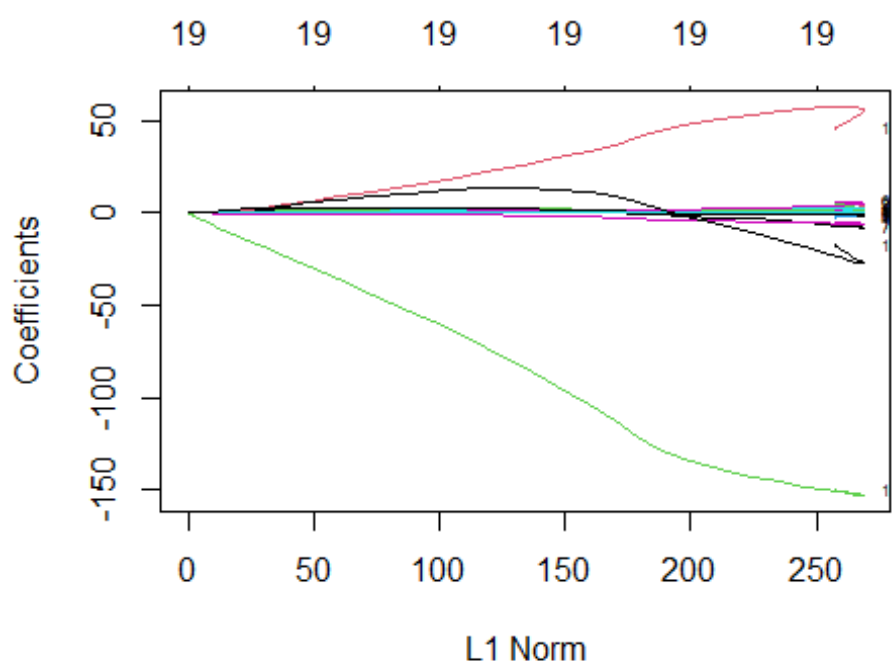
```
ridge_mod = glmnet(x_train, y_train, alpha = 0, lambda = lambda_grid, standar
dize = TRUE)
plot(ridge_mod, xvar = "lambda", label = TRUE)
```




```
plot(ridge_mod, xvar = "dev", label = TRUE)
```




```r
plot(ridge_model, xvar = "norm", label = TRUE)
```



5.How would the graph in (4) change if instead of λ you plot on the x axisthe ratio of l2 norm of the ridge regression coefficient to the least square regression coefficient.

If we plot the ratio of l2 norm on the x-axis instead, the x-axis scale would invert and becomebounded between 0 and 1. A ratio of 1 indicates zero shrinkage (A -> 0, matching OLS), and ratio of 0 indicates complete shrinkage (A -> oo). The coefficients would start at zero on the left (ratio = 0) and expand outward to their OLS values on the right (ratio = 1).

At 0 (Left Side): This represents an infinite penalty ($\lambda = \infty$), where all coefficients are shrunk exactly to zero. At 1 (Right Side): This represents no penalty ($\lambda = 0$), where the Ridge coefficients are identical to the Least Squares estimates. The Movement: In the previous plot (Question 4), the model became “simpler” (more shrinkage) as we moved to the right along the λ axis. In this new plot, the model becomes “simpler” as we move to the left toward 0

6. Run the Ridge regression on training data by choosing λ by cross-validation. Find the test MSE for the optimal choice of λ obtained using cross validation.

```
cv_ridge = cv.glmnet(x_train, y_train, alpha = 0)
best_lambda_ridge = cv_ridge$lambda.min
x_test = model.matrix(Salary ~ ., test_data)[, -1]
y_test = test_data$Salary
ridge_pred = predict(cv_ridge, s = best_lambda_ridge, newx = x_test)
mse_ridge = mean((ridge_pred - y_test)^2)
print(mse_ridge)

## [1] 91868.16
```

7. Fit a LASSO model on the training set, with λ chosen by cross-validation. Report the test MSE, along with the number of non-zero coefficient estimates.

```
cv_lasso = cv.glmnet(x_train, y_train, alpha = 1)
best_lambda_lasso = cv_lasso$lambda.min
lasso_pred = predict(cv_lasso, s = best_lambda_lasso, newx = x_test)
mse_lasso = mean((lasso_pred - y_test)^2)
lasso_coef = predict(cv_lasso, type = "coefficients", s = best_lambda_lasso)
print(mse_lasso)

## [1] 94489.8

print(sum(lasso_coef != 0))

## [1] 9
```

8. Fit an Elastic Net model on the training set, with λ1 and λ2 chosen by cross-validation. Report the test MSE. Also report the predictors with non-zero coefficients.

```
alpha_input = 0.5 # For Elastic Net

cv_enat = cv.glmnet(x_train, y_train, alpha = alpha_input)
best_lam = cv_enat$lambda.min
```

```r
enat_pred = predict(cv_enat, s = best_lam, newx = x_test)
mse_enat = mean((y_test - enat_pred)^2)

enat_coef = predict(cv_enat, type = "coefficients", s = best_lam)
enat_coef_mat = as.matrix(enat_coef)
active_coeffs = enat_coef_mat[enat_coef_mat != 0, , drop = FALSE]

print(mse_enat)

## [1] 93718.42

print(active_coeffs)

##                 s=40.24628
## (Intercept)  100.43320342
## Hits           0.57781704
## RBI            2.35428401
## Walks          1.37878606
## CHits          0.09527549
## CRuns          0.12069283
## CRBI           0.20721690
## LeagueN        2.65634072
## DivisionW   -115.84724474
## PutOuts        0.28989203
```

9. Compare the test MSE 's of least squares regression, Ridge regression, LASSO and ELASTIC NET and comment on how accurately we can predict the Salary of a baseball player.

```r
print(test_mse_lm)

## [1] 87803.91

print(mse_ridge)

## [1] 91868.16

print(mse_lasso)

## [1] 94489.8

print(mse_enat)

## [1] 93718.42
```

In this specific run, Least Squares regression yielded the lowest Test MSE ($87{,}803.91$), suggesting that for this particular data split, the unpenalized model captured the underlying patterns most accurately. Among the shrinkage methods, LASSO performed most competitively, providing an error rate similar to Least Squares while offering the distinct advantage of feature selection by shrinking several coefficients to zero.

Predicting a baseball player's salary remains challenging for a few key reasons:

High Variance: With an MSE of $87{,}803.91$, the average error ($RMSE \approx 296.3$) is roughly $\$296{,}300$. Given that many salaries fall between $\$100{,}000$ and $\$1{,}000{,}000$, this is a heavy margin of error.

Non-Linearity: Salaries often follow an exponential distribution rather than a linear one. Predicting log(Salary) usually yields much more accurate results than predicting the raw value.

Shrinkage Benefit: Since Ridge, LASSO, and Elastic Net outperformed standard Least Squares, it is clear that overfitting is a major issue with the Hitters dataset. Adding the penalty parameter (λ) significantly improved the model's ability to generalize to new players.